

Overview of Each Script:

tracking-plotbac.py: This script creates visualizations for the tracking output of various tools. You'll find the resulting tracking plots in the specific folder for each tool under the analyses folder.

reference_BW_img.py: This script starts by retrieving the output file from LABkit. It then identifies each bacterium's pixels from the LABkit output and displays them in black and white. This script is primarily used for IOU (Intersection over Union) calculations.

Jitter_remover.py: This script minimizes stage jitter effects.

delta_modified.zip: The Delta output includes the minor and major lengths of the rectangle fitting around each bacterium, aiding in determining their orientation. It removes bacteria touching the frame's border. The segmentation output is saved in both black and white (as a TIFF file) and in color (with each bacterium in a different color, in TIFF format and as an array in an NPY file). The tracking output is also stored as a TIFF image.

SuperSegger-o-modified.zip: Removes bacteria that touch the page border. The segmentation output is stored in black and white (BW) as a TIFF file and in color (each bacterium in a unique color) as a .mat file. Rename files: For image input into DeLTA or SS, a specific pattern is required. These scripts adjust the file names of raw images to align with the desired pattern for these tools, enabling their use in DeLTA and SS.

CP-omnipose & CP- post-processing/CP_Post_Processing: This suite of tools reads and post-processes output from the CP tool. It assigns a fixed ID to each bacterium for its entire lifespan and labels each family tree. It calculates metrics like birth_length, AverageLength, LifeHistory, GrowthRate, and AverageOrientation throughout their life histories, counts divisions in family trees, and reports number of bacteria at each time step.

Other specific package processing: These scripts process output from each tool, conducting post-processing and gathering data about each bacterium over time lapse. They assign consistent IDs to bacteria and label family trees. They then calculate parameters such as birth_length, AverageLength, LifeHistory, GrowthRate, and AverageOrientation over their life histories, alongside counting divisions in family trees and reporting bacterial counts at each time step.